

A PRACTICAL FIELD GUIDE / 2026

AI, LLMs & AGENTIC SYSTEMS

A clear, practical introduction to language models, AI agents, and systems that can plan and act.

THE SHORT VERSION

An LLM generates language. An AI agent uses a model with tools and a control loop. An agentic system gives that loop a goal, bounded autonomy, and ways to check its work.

01 UNDERSTAND

Build a shared vocabulary for AI, machine learning, and LLMs.

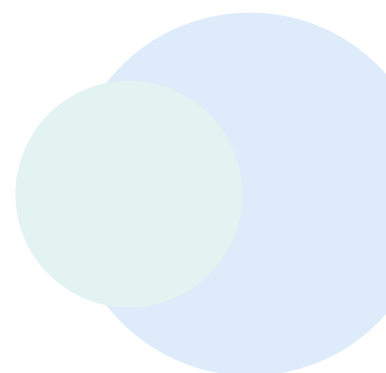
02 DESIGN

See how models, tools, memory, and retrieval fit together.

03 DELIVER

Choose guardrails and evaluations before adding autonomy.

Designed for learners, builders, and teams evaluating LLM-powered products.



CONTENTS

A map of the guide

Eight pages, moving from fundamentals to an implementable document-assistant architecture.

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01 Foundations	AI, machine learning, LLMs, tokens, and context	03
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HOW TO USE THIS GUIDE

Read the first four sections for the conceptual model. Use the PDF assistant blueprint as a concrete example of retrieval-augmented generation (RAG). Use the final checklist before exposing an agent to real users, private data, or consequential actions.

A useful system map

MODEL	CONTROL	CAPABILITIES	EVIDENCE
Interprets input and proposes output	Application code sets limits and flow	Tools perform bounded operations	Retrieved sources and tests support claims

A NOTE ON TERMINOLOGY

There is no single universally enforced boundary between an AI agent and an agentic system. In this guide, an agent is a model-driven system that can use tools in a loop; agentic describes stronger goal-directed behavior and autonomy.

01 / FOUNDATIONS

From AI to LLMs

Artificial intelligence is the broad field of building systems that perform tasks associated with perception, prediction, reasoning, or generation. Most current language assistants are specialized systems, not general human-level intelligence.

A compact vocabulary

TERM	MEANING	EXAMPLE
Artificial intelligence	A broad label for machine-based systems that produce predictions, recommendations, or actions.	Speech recognition or route planning
Machine learning	Methods that fit patterns from data rather than relying only on hand-written rules.	A classifier trained on labeled examples
Deep learning	Machine learning with multi-layer neural networks that learn representations.	Vision, speech, and language models
Generative AI	Models that create new text, images, audio, video, or code from learned patterns.	Drafting a summary from a prompt
LLM	A large language model trained to process and generate token sequences.	Answering a question or drafting code

How an LLM responds

- 1. **Tokenize:** split text into model-specific units called tokens; a token can be a word, part of a word, or symbol.
- 2. **Represent context:** map tokens to numerical representations and process relationships across the input sequence.
- 3. **Generate:** repeatedly predict a likely next token, subject to the prompt, model settings, and context window.
- 4. **Decode:** turn the generated token sequence back into text or another supported output format.

Terms that affect application design

CONTEXT WINDOW	The input and output token budget available to a model call.
EMBEDDING	A vector representation used to compare semantic similarity between items.
INFERENCE	Running a trained model to produce an output.
HALLUCINATION	A plausible-sounding claim that is unsupported or incorrect.

DESIGN CONSEQUENCE
Fluent output is not proof of truth. Ground important answers in trusted data, expose uncertainty, and verify behavior with tests.

02 / AI AGENTS

A model with a control loop

An AI agent combines a model with instructions, state, and tools. The surrounding application decides what the model may do, validates its requests, and determines when the task is finished.

The main components

COMPONENT	RESPONSIBILITY	DESIGN QUESTION
Model	Interpret the request and choose a next step.	Which model meets quality, latency, and cost needs?
Instructions	Define role, objective, constraints, and output expectations.	Are rules explicit and testable?
Tools	Expose bounded operations such as search or calculations.	What permissions and input validation does each tool need?
State	Carry conversation context and task progress.	What is transient, persistent, or sensitive?
Controller	Validate calls, apply policies, manage retries, and stop.	What is the maximum number of steps or spend?

The observe-act loop

1 RECEIVE Goal, context, constraints	2 DECIDE Model proposes an answer or tool call	3 ACT Controller validates and runs it	4 OBSERVE Return result; continue or stop
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Chatbot versus tool-using agent

BASIC CHATBOT	TOOL-USING AGENT
Usually maps one request to one response.	May take several bounded steps and call tools.
Uses supplied context and conversation history.	Can retrieve information or perform permitted actions.
Often easier to predict and evaluate.	Needs step limits, tool checks, and failure handling.

KEEP TOOLS NARROW
A search tool should search. A payment tool should require authorization and enforce limits in code. Never rely on prompt wording alone to enforce permissions.

03 / AGENTIC AI

More agency, more responsibility

Agentic AI describes systems designed to pursue goals through planning, tool use, feedback, and multiple steps. The term is used inconsistently, so evaluate the actual capabilities and permissions rather than the label.

A practical autonomy spectrum

LEVEL	SYSTEM BEHAVIOR	HUMAN CONTROL
0 Assist	Generates a suggestion or draft.	Person decides and performs the action.
1 Retrieve	Searches approved sources and returns evidence.	Person reviews the result.
2 Bounded act	Executes a reversible, low-risk action within limits.	Policy checks, logging, and undo are available.
3 Multi-step	Plans and performs several actions toward a goal.	Checkpoints and approvals gate consequential steps.

Patterns and their tradeoffs

PATTERN	GOOD FIT	WATCH FOR
Retrieval-augmented generation	Answers that need current or private reference material.	Poor retrieval can make a grounded prompt look trustworthy while missing key evidence.
Plan then execute	Tasks with clear stages and a verifiable end state.	Plans can be wrong; validate each action, not just the plan.
Tool routing	Choosing among a small set of well-defined capabilities.	Ambiguous tool descriptions and permissive APIs cause misfires.
Multi-agent workflow	Distinct roles that benefit from independent review.	Coordination adds latency, cost, and more failure paths.

When not to add autonomy

Use a fixed workflow when steps are known. Use a retrieval pipeline when the main need is finding evidence. Add an agent loop only when choosing the next step dynamically creates meaningful value and can be kept within safe boundaries.

AUTONOMY SHOULD BE EARNED

Start with read-only actions, narrow the task, add explicit stop conditions, and expand permissions only after evaluations show reliable behavior.

04 / LLM LANDSCAPE

Choose a model for the job

There is no best model in isolation. Compare candidate models on representative tasks, with the same prompts, tools, constraints, and evaluation set you expect in production.

Common model categories

CATEGORY	TYPICAL STRENGTH	TRADEOFF TO TEST
General-purpose LLM	Broad language, summarization, and instruction following.	Cost, latency, and consistency on your domain.
Reasoning-focused model	Complex decomposition and multi-step problem solving.	Latency, token use, and whether extra reasoning improves measured results.
Small or edge model	Low latency, lower serving cost, and local deployment options.	Capacity limits, hardware, and quality on long or difficult inputs.
Multimodal model	Combines text with image, audio, or video input/output.	Supported formats, extraction quality, and data handling.
Embedding model	Maps text or other content to vectors for retrieval.	Domain relevance, language support, dimensions, and index compatibility.

Examples and selection criteria

Widely known model families include GPT, Claude, Gemini, Llama, Qwen, and DeepSeek. Each family includes multiple models and deployment choices; names, limits, and capabilities change frequently. Verify current provider documentation and run your own benchmark before choosing.

CHECK	ASK
Quality	Does it answer your real questions correctly and cite evidence?
Latency and cost	What is end-to-end time and cost per successful task?
Privacy and deployment	Where are prompts processed, retained, and logged?
Reliability	Does output stay structured and follow tool constraints?
Operations	Can you monitor, version, and switch providers if needed?

RAG, fine-tuning, or longer context?

APPROACH	USE WHEN
RAG	Answers need traceable facts from a changing knowledge base.
Fine-tuning	You need consistent behavior or style learned from examples, not a substitute for frequently updated facts.
Long context	The relevant source fits the model context and simpler retrieval is not necessary.

05 / PDF ASSISTANT BLUEPRINT

From uploaded PDF to sourced answer

A document assistant combines an ingestion path with a question-answering path. Keep both explicit so you can inspect what was stored and why a source was retrieved.

Two connected flows

INGEST A DOCUMENT	ANSWER A QUESTION
1. Validate PDF type and size. 2. Extract text and page metadata. 3. Split text into overlapping chunks. 4. Embed chunks and upsert IDs into Chroma.	1. Embed the user question. 2. Retrieve the most relevant chunks. 3. Give question and excerpts to the LLM. 4. Return answer with file and page citations.

Metadata that makes retrieval useful

FIELD	WHY KEEP IT
Document ID and content hash	Stable updates and duplicate detection.
Original filename	Readable citations and troubleshooting.
Page number	Let a person verify an answer in the source.
Chunk index and version	Trace retrieval to an ingestion run.
Access scope	Prevent retrieving documents a user is not allowed to see.

Retrieval quality matters

Chunk boundaries should preserve meaning. Too-small chunks lose context; too-large chunks dilute relevance. Tune chunk size and overlap on representative PDFs, then inspect retrieved passages for real questions. Consider OCR for scanned pages, and report extraction failures instead of silently indexing empty text.

MEMORY IS NOT A KNOWLEDGE BASE

Conversation memory preserves useful interaction state. A vector index retrieves reference material. Keep them separate, define retention rules, and do not store sensitive data without a clear need.

06 / RESPONSIBLE DELIVERY

Ship with evidence and limits

A useful prototype answers a question. A dependable system also shows its sources, handles failure, protects data, and has a clear boundary for actions.

Risk and control checklist

RISK	PRACTICAL CONTROL
Unsupported answer	Test against known questions; retrieve sources; allow an explicit 'not found' response.
Prompt injection in a PDF	Treat retrieved text as untrusted data, not instructions. Restrict available tools.
Unsafe tool action	Validate arguments in code; use allowlists, approvals, and least privilege.
Sensitive data exposure	Minimize collected data; review provider retention; redact and set deletion policies.
Silent failure	Log request IDs, retrieved source IDs, tool outcomes, latency, and errors without logging secrets.

Evaluate the whole system

MEASURE	SAMPLE QUESTION
Answer correctness	Is the response factually supported by the source?
Retrieval quality	Did the top results include the passage needed to answer?
Citation quality	Do cited file and page details point to the evidence used?
Safety and refusal	Does the system avoid unauthorized actions and admit missing evidence?
Operations	What are latency, cost, error rate, and recovery behavior?

Before launch

- [] Define allowed tasks and prohibited actions. [] Test normal, ambiguous, and adversarial inputs.
- [] Require approval for consequential actions. [] Check data retention and provider settings.
- [] Monitor failures and provide a human escalation path.

Further reading

Vaswani et al., *Attention Is All You Need* (2017), arxiv.org/abs/1706.03762 | Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks* (2020), arxiv.org/abs/2005.11401 | NIST AI Risk Management Framework, nist.gov/itl/ai-risk-management-framework | OWASP Top 10 for LLM Applications, genai.owasp.org

TAKEAWAY

Use the least autonomy that solves the problem. Measure the complete workflow, keep a human in control of consequential decisions, and make evidence easy to inspect.

Educational overview; model behavior, provider terms, and product capabilities change. Verify current documentation before deployment.